OPTIMAL AMBULANCE POSITIONING FOR ROAD ACCIDENTS WITH DEEP EMBEDDED CLUSTERING

ABSTRACT

The number of casualties and fatalities brought on by road accidents is one of the most significant concerns in the modern world. Instead of dispatching ambulances only at the time of demand, pre-positioning them can reduce the response time and provide prompt medical attention. Deep learning techniques hold great potential and have proven to be essential for problem-solving and decision-making in the field of healthcare services. This study introduces a deep-embedded clustering-based approach to predict optimal locations for ambulance positing. Various factors and patterns in a geographical region greatly influence the occurrence of road crashes, hence understanding such relationships while model building is crucial. The present study also emphasizes the need of preserving such patterns during model building to

ensure real-time results and implements them with the help of another deep-learning-based model, Cat2Vec. The proposed framework is also compared with traditional clustering algorithms like K-means, GMM, and Agglomerative clustering. Moreover, to calculate response time and distance in real time, a novel scoring function has also been introduced for the performance evaluation of various algorithms. The proposed

ambulance-positing system exhibits remarkable performance, achieving an accuracy of 95% with k-fold cross-validation and a novel distance score of 7.581 proving the use of the proposed approach is better than all the other traditional algorithms used.

INTRODUCTION

Today one of the leading causes of death worldwide among children and adults is road accidents. The injuries caused by these fatal accidents cause considerable economic and personal losses to individuals, their families, and the country. An estimated 1.3 million individuals each year die as a result of road accidents. Between 20 and 50 million individuals experience non-fatal injuries, with many of them becoming disabled as a result [1]. The ever-increasing growth in the number of automobiles is certain to have some negative. consequences, the most likely of which is an increase in the frequency of fatal road accidents in densely populated places, resulting in a huge burden on the urban infrastructure. It is dreaded that if we fail to take definitive precautionary measures to overcome these statistics, then road accidents will take over as the fifth major cause of death by 2030. Despite these fatal consequences, this problem receives scanty attention and there is a lack of developing systematic methods to improve road safety.

Studies show that over 90 percent of the global traffic accidents occur in medium to lower income countries such as Kenya [2], [3] which is one such example as more than a thousand fatalities occur due to road crashes consisting of a mean of 7 out of 35 casualties each day [4]. Majority of these deaths and severe injuries occur to the population of 15-59 years who are also the economically active citizens of the country, reducing the economic activity of the country.. Kenya, as a country ranking in the range of lower-middle income, has seen an increase in regional trade deals over the past decade. Based on the reports from National Transport Safety Authority (NTSA), which is the agency responsible for transportation in Kenya, a record of 5186 minor injuries, 6938 major injuries and 3572 deaths was concluded in 2019.

The number of injuries, and fatalities brought on by these deadly accidents can be decreased if preventive measures are taken, the most crucial of which are prompt medical attention for accident victims, information about the precise situation to the aid personnel, and accurate data analysis considering every single factor to diagnose and predict the accident-prone zones in a city. The delay in the arrival of an ambulance has a significant impact on human life especially in the case of emergency response pertaining to road accidents [5]. If the ambulance fails to reach the crash site in the critical hour, casualties may increase, therefore making each second very significant to human life. In every big metropolis, choosing the best places to place emergency responders throughout the day as they wait to be summoned is essential due to the dense traffic patterns and the city's distinctive layout. Monitoring and controlling these killer accidents is even more difficult due to the lack of expertise in stationing emergency response systems. Therefore, the prompt, automated, and timely positioning of ambulances can aid the first responders and doctors by reducing the effort required on their end and enabling earlier treatment decisions.

In this modern era technologies like machine learning and deep learning have always proven to be an emphatic and prevalent approach to decision-making, especially in the field of medical services. The advent of these technologies has been helpful in various road safety problems and their usefulness can also be found in our problem statement. Healing all patients and eliminating casualties in road accidents is needed of the hour as the end goal of improvement in Health Care Output (HCO). This paper considers the optimal positioning of the ambulance (paramedic help) as a clustering problem, since clustering algorithms ensure optimal locations based on distance metrics and, the coordinates of each centroid are the means of the coordinates of the objects in the cluster [6]. Traditional machine learning approaches such as k-

means clustering, PAM clustering, Agglomerative clustering, etc. are not adequately versed and practical in all kinds of clustering problems [7]. This causes a novel deep learning-based technique to develop, which is an exercisable way to enhance this process' performance.

In this study, we propose a novel clustering-based approach utilizing Deep Embedded Clustering with Auto encoder (DEC-AE) to address the problem of optimal ambulance positioning in a city. Unlike traditional clustering methods, the DEC-AE method offers a comprehensive framework that combines deep learning, clustering, and auto encoder techniques [23] to optimize ambulance positioning strategies. By reconstructing the input data from the learned latent representations, DEC can effectively capture the essential features and dimensions that contribute to the clustering process. Furthermore, DEC employs a joint optimization objective [24] that integrates clustering assignments and feature learning. This joint optimization facilitates the enhancement of cluster separate ability and the generation of compact and well- separated clusters in the latent space. DEC-AC combines deep learning and adaptive clustering [25] to provide an effective solution for clustering problems. It leverages deep neural networks to learn meaningful feature representations and adaptively determines the number of clusters based on the data distribution.

Additionally, DEC is scalable and can handle large-scale datasets, making it suitable for real-world applications with high-dimensional and complex data. This enables a more accurate and nuanced understanding of the factors influencing optimal ambulance positioning. The DEC-AE approach also incorporates clustering algorithms, facilitating the identification of clusters or groups of similar patterns within the data [25]. This allows for the identification of hotspot areas with higher accident probabilities or specific risk profiles, aiding in the strategic placement of ambulances to minimize response times and maximize coverage. Additionally, this method has the potential to accommodate diverse data sources, including traffic accident data, road segment characteristics, weather conditions, and other relevant factors. By considering multiple data dimensions, the approach can provide a holistic view of the problem, enhancing the precision and effectiveness of ambulance positioning strategies.

The dataset includes information on traffic accidents that occurred, road segment information, and weather details of Nairobi, Kenya. Performing Exploratory Data Analysis on the dataset of the road surveys, and weather dataset, the paper identifies possible features and attributes affecting the accidents and patterns of risk across the city. To preserve such relationships and patterns of the data we apply a deep learning-based embedding approach called Cat2Vec while converting categorical attributes in the data pre-processing stage. To validate the predicted locations using DEC, the distance from that crash site to the nearest ambulance locations predicted is calculated using a novel Distance Scoring Algorithm. For further evaluation of the algorithm, different clustering metrics have been used and compared with other traditional clustering algorithms.

The proposed methodology in this paper addresses the following aspects:

- Exploratory Data Analysis (EDA) is performed on the real-time accident dataset through which the potential features and attributes which contribute towards accidents and patterns across the city are identified.
- A clustering-based approach using Deep Embedded Clustering (DEC) is developed to identify optimal ambulance positioning locations across Nairobi while preserving the feature relationships and patterns using Cat2Vec deep learning-based embedding technique which facilitates more accurate clustering.

A novel Distance Scoring method is developed to validate the DEC model which calculates the distance between crash-site and the nearest predicted ambulance location, thus providing a quantitative measure of effectiveness.

• The performance of the proposed framework is then evaluated with and without the feature selection techniques and compared with existing clustering methods using various clustering metrics, which further validates the effectiveness of the DEC model.

The rest of this paper is organized as follows. Sect. II is a detailed review about related works for the problem statement. In Section III the methodologies and materials used are described, Section IV discusses the experiments performed and results obtained from the proposed framework. The paper delivers the discussion and future scope in Section V.

SYSTEM ANALYSIS

EXISTING SYSTEM

Assi et al. [8] and Xiong et al. [27] proposed Machine learning models for predicting accident vs non-accident patterns in crash sites using Gaussian Mixture models and SVM. They further predicted the severity of the crash injuries by clustering the crashes using fuzzy c-means, Feed Forward Neural Networks and SVM. Data analysis was performed to identify the features from the crash sites to provide inputs to the ML models. The models were evaluated using accuracy, sensitivity and precision. On comparing the models, fuzzy c-means algorithm provided greater accuracy than traditional k-means clustering and SVM models.

Ghandour et al. [9] and Tiwari et al. [10] developed an approach that uses a machine learning hybrid ensemble classifier derived from decision trees and MSO algorithm to identify risk factors that contribute to fatal road accidents. They utilized the Lebanese Road accident platform (LARP) dataset consisting of 8482 accidents and the fatalities occurred in the accidents. To evaluate the impact of the factors causing casualties in a road accident, they performed sensitivity analysis of the attributes. From the selected variables, seven of nine showed significant association to casualties. To evaluate the model performances the metrics used were F1 score,
precision, AUC-PR curve and Cohen's Kappa. Granberg et al. [11] developed a simulation based predictive model to gauge the emergency ambulance demand in an
area using multivariate regression model.

The data used for their genetic regression algorithm was collected from census survey of the year 2005 consisting of 2076 small areas. For each location, a distance matrix was developed and used as inputs to the genetic algorithm in order to identify 35 probable ambulance locations using R-statistics software. The proposed model favored a significant R2 value of 0.71 with multiple coefficients. This particular distance matrix based approach provided better results on comparing with the models using nave forecasting techniques. Clustering in machine learning has been explored pertaining feature selection techniques [12], [13], [14], distance functions and cluster validations.

A derivative of the popular clustering methods are K-means and Gaussian mixture models. Even though the approach using distance functions was developed earlier, its application and popularity
are limited by high dimensionality and dataset space. Cao et al. [16] and Moriya et al. [17] proposed approaches based on batch clustering, fuzzy c-means clustering, and K-means clustering with an FNMF matrix for clustering illustrating the correlation patterns of the initial data points. In both the approaches, the correlation among the crash locations are calculated followed by clustering of the said crash locations based on the factors responsible for the accidents.

Alkheder et al. [18] proposed an approach using decision tree classifier, MLP and Naïve Bayes to identify the significant attributes that impact the prediction of the severity of a road accident. On comparing different models, it was concluded that the decision tree classifier provided a better classification accuracy of 0.08218. The attributes such as year of accident, age, nationality, gender and the type of accident were more significant in determining the accident severity.

Hashmienejad et al. [19] utilized decision trees and genetic algorithms to develop a prediction model for predicting the severity of the road accidents. The set of rules induced from the genetic algorithm approach were further provided as input to the decision tree models namely CART, C4.5 and ID3, also validated using the test dataset. The method employed provided an accuracy of 0.8820, recall of 0.889, f-measure of 0.887 and a precision of 0.885 which was better than the alternative methods used namely ANN, SVM, KNN and Naïve bayes.

Ghosh et al. [20] and Sasaki et al. [21] employed Bayesian networks (BN) approaches to develop models based on the relationship of the attributes which were represented as probability distributions. Bayesian Networks are used for identifying the factors responsible for road accidents and also predict the severity of the accidents. These papers also evaluated the sensitivity and specificity of the models
along with MAE and RMSE to assess the performance of BSVR. Taamneh et al. [22] utilized Artificial Neural Networks (ANN) along with K-means to predict the severity of road accidents. Other machine learning models were also used to compare the accuracy of the proposed ANN model, concluding that the proposed model provided a higher accuracy of 0.746.

Dizaji et al. [23] and Tian et al. [24] used Auto-encoders to reduce the dimensionality for obtaining the features having higher impact on clustering. Following the dimensionality

reduction, Kmeans is employed for clustering the features in groups. The approach employs auto encoders to obtain a representational structures of the locations, then neglects the decoder part to obtain a smooth model and finally adds
the Kmeans layer over the encoder layer to obtain the final model. Although this approach uses a deep neural network for mapping the data points in feature space prior to the feature selection and cluster formation, the goals of these two separate processes are not optimized together.

Alqahtani et al. [25] proposed an approach using embedded clustering layer in deep auto-encoders. When compared to traditional clustering methods, this technique learns the feature representations and assigns clusters concurrently through deep auto encoders. During the optimization phase, the cluster centers are reassigned to all the data points which are accident locations in this particular problem. Following this, the cluster centers are updated iteratively in order to obtain the final stable clusters and better optimized performance.

DISADVANTAGES

• The complexity of data: Most of the existing machine learning models must be able to accurately interpret large and complex datasets to detect Ambulance Positioning.

• Data availability: Most machine learning models require large amounts of data to create accurate predictions. If data is unavailable in sufficient quantities, then model accuracy may suffer.

• Incorrect labeling: The existing machine learning models are only as accurate as the data trained using the input dataset. If the data has been incorrectly labeled, the model cannot make accurate predictions.

PROPOSED SYSTEM

The dataset includes information on traffic accidents that occurred, road segment information, and weather details of Nairobi, Kenya. Performing Exploratory Data Analysis on the dataset of the road surveys, and weather dataset, the paper identifies possible features and attributes affecting the accidents and patterns of risk across the city. To preserve such relationships and patterns of the data we apply a deep learning-based embedding approach called Cat2Vec while converting categorical attributes in the data pre-processing stage. To validate the predicted locations using DEC, the distance from that crash site to the nearest ambulance locations predicted is calculated using a novel Distance Scoring Algorithm. For further evaluation of the algorithm, different clustering metrics have been used and compared with other traditional clustering algorithms.

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and patterns using Cat2Vec deep learning-based embedding technique which facilitates more accurate clustering.

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ADVANTAGES

The proposed system proposes an approach (optimal ambulance positioning framework) for the automatic placement of paramedic help using Deep Embedded Clustering (DEC). To ameliorate the classification accuracy, this study utilizes Cat2vec(a deep learning-based model) to represent high cardinality categorical variables using low-dimension embedding while preserving the relationship and patterns obtained through exploratory data analysis between each of the categories. This study employs K-fold cross-validation for dividing the dataset into training and test sets.

SYSTEM REQUIREMENTS

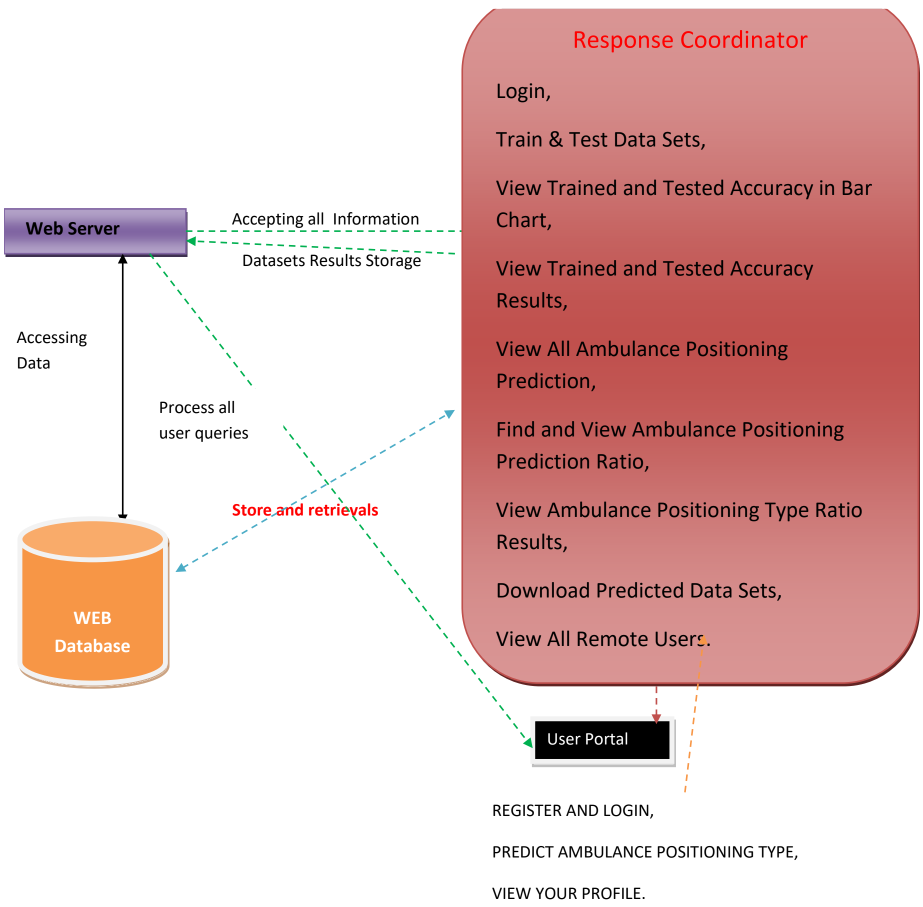
HARDWARE REQUIREMENTS

- ➢ Processor - Pentium –IV
- ➢ RAM - 4 GB (min)
- ➢ Hard Disk - 20 GB
- ➢ Key Board - Standard Windows Keyboard
- ➢ Mouse - Two or Three Button Mouse
- ➢ Monitor - SVGA

SOFTWARE REQUIREMENTS

- ❖ Operating system : Windows 7 Ultimate.
- ❖ Coding Language : Python.
- ❖ Front-End : Python.
- ❖ Back-End : Django-ORM
- ❖ Designing : Html, css, javascript.
- ❖ Data Base : MySQL (WAMP Server).

SYSTEM ARCHITECTURE



MODULES

- Response Coordinator
- User Portal